Zhaoyi Xu

Homepage | Zhaoyi Xu | zhaoyix@umich.edu | xu-zhaoyi | Zhaoyi Xu

Research Interests

- Bioengineered human pluripotent stem cell (hPSC) organoids
- Synthetic human development and disease modeling
- Integration of *in vitro* experiments with *in silico* models

Education

University of Michigan, Ann Arbor 2026
Ph.D. in Mechanical Engineering, GPA: 4.0 / 4.0
Advisor: Prof. Jianping Fu
Thesis:
hPSC-Based Modeling of Human Embryonic Development via Acoustic and Microfluidic Approaches

University of Toronto 2021
M.A.Sc. in Mechanical Engineering, GPA: 3.9 / 4.0
Advisor: Prof. Xinyu Liu
Thesis: An Improved Approach for Optogenetic Locomotion Control of Caenorhabditis Elegans

Hong Kong University of Science and Technology 2019
B.Eng. in Mechanical Engineering, Dean's List, First Class Honours
Exchange student, National University of Singapore (2018)

Selected Research Experience

Synthetic 3D hPSC Models on Peri-Implantation Development 2021 - 2026
Supervisor: Prof. Jianping Fu *University of Michigan*

- Developed microfluidic-based biomimetic substrates for 3D hPSC organoid culture to study early human embryo development.
- Led the execution of three projects, resulting in publications [1] [9] [11].

Neural Differentiation of hPSCs via Acoustic Tweezing Cytometry 2022 - 2024
Supervisors: Prof. Jianping Fu, Prof. Cheri X. Deng *University of Michigan*

- Utilized acoustic tweezing cytometry technology to study mechanosensitive properties of hPSCs
- Achieved scalable and reproducible production of functional motor neurons for medical applications.
- Core member of NIH R01 grant GM143297, resulting in publications [4] [10].

Improvement of Optogenetic Locomotion Control of C. elegans 2019 - 2021
Supervisor: Prof. Xinyu Liu *University of Toronto*

- Facilitated the development of a living soft microrobot - RoboWorm (optogenetic C. elegans controllable by microfluidic tools), resulted in a highly-cited publication [5].
- Improved protocol success rate from 7.1% to 80.77% by creating a new transgenic worm strain and enhancing control system robustness.

Publications

Journals

[1] **Zhaoyi Xu**, Weiping Li, Xufeng Xue, Nicholas G. Schott, Ruxin Yang, Shiyu Sun, Jeyoon Bok, Jan P. Stegemann, Jianping Fu, and Cheri X. Deng, "Acoustic cell patterning reveals geometry- and substrate-dependent vasculogenesis and human embryo model development," *Nature Communications*, 2026. DOI: 10.1038/s41467-026-76976-5.

[2] Tao Hong, **Zhaoyi Xu**, Se Young Chun, Luis Hernandez-Garcia, and Jeffrey A. Fessler, "Convergent complex quasi-Newton proximal methods for gradient-driven denoisers in compressed sensing MRI reconstruction," *IEEE Trans. on Computational Imaging*, vol. 11, pp. 1534–1547, 2025. DOI: 10.1109/TCI.2025.3625052.

[3] Tao Hong, **Zhaoyi Xu**, Jason Hu, and Jeffrey A. Fessler, "Using randomized Nyström preconditioners to accelerate variational image reconstruction," *IEEE Trans. on Computational Imaging*, vol. 11, pp. 1630–1643, 2025. DOI: 10.1109/TCI.2025.3622903.

[4] **Zhaoyi Xu**, Shiying Liu, Xufeng Xue, Weiping Li, Jianping Fu, and Cheri X Deng, "Microbubble behaviors in acoustic tweezing cytometry regulates embryonic stem cell differentiation," *Scientific Reports*, vol. 13, no. 18030, 2023. DOI: 10.1038/s41598-023-45397-5.

[5] Xianke Dong, Sina Kheiri, Yangning Lu, **Zhaoyi Xu**, Mei Zhen, and Xinyu Liu, "Towards a living soft microrobot: Optogenetic locomotion control of caenorhabditis elegans," *Science Robotics*, vol. 6, no. 55, eabe3950, 2021. DOI: 10.1126/scirobotics.abe3950.

[6] Mingcan Liu, **Zhaoyi Xu**, Jing Jie Ong, Jian Zhu, and Wen Feng Lu, "An earthworm-like soft robot with integration of single pneumatic actuator and cellular structures for peristaltic motion," *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, pp. 7840–7845, 2020. DOI: 10.1109/IROS45743.2020.9341166.

[7] Sihan Huang, **Zhaoyi Xu**, Guoxin Wang, Cong Zeng, and Yan Yan, "Reconfigurable machine tools design for multi-part families," *The International Journal of Advanced Manufacturing Technology*, vol. 105, no. 1, pp. 813–829, 2019. DOI: 10.1007/s00170-019-04236-6.

Conferences

[8] **Zhaoyi Xu**, Weiping Li, Jianping Fu, and Cheri X Deng, "Acoustic cell patterning reveals geometry- and substrate-dependent vasculogenesis and human embryo model development," Society of Engineering Science (SES) Annual Meeting, West Lafayette, IN, US, 2026, **(Talk)**.

[9] **Zhaoyi Xu**, Weiping Li, Jianping Fu, and Cheri X Deng, "Controlled development of a human epiblast model using acoustic patterning," 8th Bioengineering and Translational Medicine Conference by AIChE, San Diego, US, 2024.

[10] **Zhaoyi Xu**, Weiping Li, Jianping Fu, and Cheri X Deng, "Formation and development of a human epiblast model in biomimetic implantation-like niche through acoustic hpscs patterning," The Great Lakes Developmental Biology Meeting, Toronto, Canada, 2023.

[11] **Zhaoyi Xu**, Xufeng Xue, and Jianping Fu, "Probing mechanobiology of human neural crest cell development using a patterned microfluidic neural tube model," Keystone Symposia on Molecular and Cellular Biology, Keystone, CO, United States, 2022.

[12] Yafeng Han, **Zhaoyi Xu**, and Wen Feng Lu, "Design of conforming surface for human skin at highly-stretched joint areas with additive manufacturing," 5th International Conference on Additive Manufacturing and Bio-Manufacturing (ICAM-BM), Beijing, China, 2018.

Honors and Awards

2026	UMich Rackham Graduate Student Research Grant	USD 3,000
2024	8th Bioengineering and Translational Medicine Conference	Young Investigator Award
2021-24	UMich Rackham Travel Grant	USD 3000
2021	UMich ME Department Fellowship	USD 89,775
2019-21	UofT MIE Department Fellowship	Full Scholarship CAD 68,000
2021	UofT Barbara and Frank Milligan Fellowships	CAD 10,000
2020	UofT MIE Endowed Fellowship	CAD 3,000
2019	HKUST President's Cup	Gold Award
2018	HKSAR Government Talent Development Scholarship	HKD 10,000
2016	HKUST Robot Design Contest	Champion
2013	China National Awarding Program of Future Scientists	Top 100
2009	China Arts Grade Examination - Piano, Grade 10	Top Level

Teaching Experience

ME 335 Heat Transfer Winter 2026
Graduate Student Instructor - 93 students *University of Michigan*

- Facilitated and coordinated the delivery of lectures.
- Designed and held course recitation sessions.

ME 495 Laboratory II Fall 2025
Graduate Student Instructor - 24 students *University of Michigan*

- Developed lab sessions for a senior-level lab course.
- Closely mentored students in both theoretical knowledge and practical experimental skills.
- Topics included heat transfer, mechanical dynamics, and AFM-based nanoscale metrology.

MIE 342 Circuits with Applications to Mechanical Engineering Systems Fall 2020
Teaching Assistant - 130 students *University of Toronto*

- Led a four-person TA team to support a lab course on circuits.
- Co-developed online laboratory tools for remote instruction during COVID-19, enabling virtual circuit assembly and experimental data acquisition.

Professional Development

U-M Graduate Teacher Certificate Program 2025-2026
University of Michigan, Rackham Graduate School and Center for Research on Learning and Teaching

- Completed pedagogy training, two semesters of GSI teaching with instructional consultation, faculty teaching mentorship, and a reviewed teaching philosophy statement.

Rackham–CRLT Preparing Future Faculty Seminar 2026
University of Michigan, Center for Research on Learning and Teaching (5-week program)

- Developed a course syllabus and teaching statement; explored faculty roles across institution types.

Student Mentoring

Linyuan Li Jun. 2025 – Sep. 2025
B.S. Student in Biochemistry *University of Michigan*

- Mechanical regulation of zebrafish PSM cells using geometrically modulated elastomeric substrates.

Ruxin Yang Aug. 2024 – Sep. 2025
M.S. Student in Biomedical Engineering *University of Michigan*

- Acoustic cell patterning.

Professional Service

Journal Reviewers: Nature, Cell Stem Cell

Skills

Biology: Cell Culture • Immunostaining • Microscopy • qPCR • Single-cell RNA Sequencing

Manufacturing: Microfluidics • Microfabrication • 3D Printing/Bioprinting

Quantitative Analysis: Python • R • Matlab

Illustration: ImageJ • Adobe Photoshop • Adobe Illustrator • Procreate

Languages: English • Mandarin • Cantonese